Metric-Based Emotional NPCs using Shared Fuzzy Logic in Top-Down Games

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Abstract - Traditional game NPCs rely on predictable AI, which interrupts player immersion. Although adding emotion-driven behavior makes them more realistic, it usually leads to significant performance issues when scaled. This paper presents a Shared Fuzzy Logic architecture to address this problem. By centralizing the decision-making process, this system allows large groups of NPCs to show dynamic, realistic behaviors without compromising the game engine's memory or real-time performance.

[1] Introduction

In modern video games, Non-Player Characters (NPCs) are crucial for creating immersive and engaging gameplay [1]. Each NPC takes on various roles, like background characters that make the game world feel more vibrant. However, many traditional NPCs have simplified and predictable decision-making systems. This can make the gameplay experience less exciting and diminish overall player engagement.

To overcome this issue, artificial intelligence techniques, like fuzzy logic, can improve NPC decision-making [2][3]. Fuzzy logic is chosen for its computational efficiency. A fuzzy logic system allows NPCs to assess, collect, and evaluate players' conditions and behaviors more effectively. As a result, NPC behavior becomes more dynamic and responsive to players.

This research explores the use of artificial intelligence algorithms, especially shared fuzzy logic, to make the game more unpredictable, enhance player satisfaction, and boost engagement. It will also describe the scenarios where fuzzy logic is applied.

[2] Literature Review

Non-Player Characters (NPCs) have historically depended on rigid, deterministic systems. Despite advancements in the rendering pipeline and physics engine, NPCs still stick to predefined rules or paths [1]. For decades, the industry has relied on two main frameworks for NPC decision-making:

- FSMs work by assigning NPCs a limited set of states, such as Patrol, Chase, and Attack [1][4][5]. Transitions between these states follow strict, hard-coded rules.

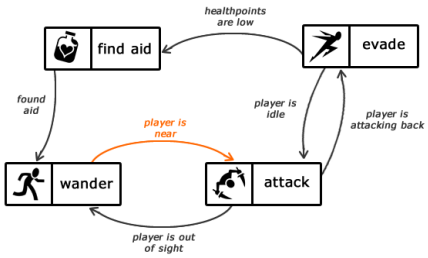

Fig 1. Standard Finite State Machine (FSM) architecture for NPC behavior.

Transitions between these states follow strict, hard-coded rules. While FSMs are cost-effective in terms of computation, they do not scale well over time. As new behaviors are added, the number of required transitions grows quickly, leading to a fragile framework often called a "spaghetti state machine" [6][7][8].

- Behavior Trees (BTs) offer a hierarchical, tree-like framework of nodes that process logic from top to bottom and left to right, creating a more reusable structure [1][4][5].

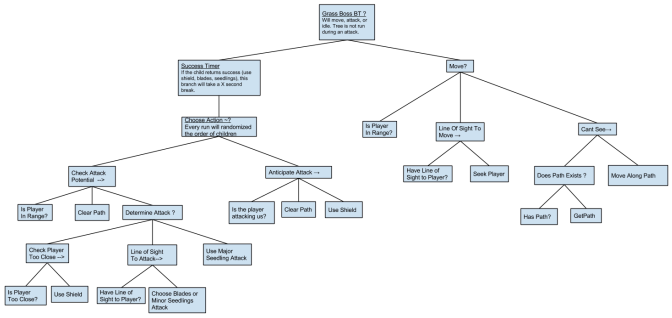

Fig 2. Hierarchical node structure of a Behavior Tree

While these decision-making frameworks are efficient and easy for developers to create and debug, they follow strict Boolean logic. This means NPCs assess their environment using fixed thresholds, leading to predictable interactions.

The dependence on these logical and strict frameworks has created a gap in player immersion. When NPCs respond predictably to complex and changing situations, the illusion of a living world shatters. According to Yannakakis and Togelius, modern AI techniques enable developers to create NPCs that feel more believable and human-like [1].

Further research, including work by Setiono et al. (2021), shows that incorporating emotional computing into game systems to process emotional interactions significantly enhances player immersion compared to traditional binary logic [9][10][11]. However, implementing such models into fast-paced video games presents practical challenges, as keeping track of detailed emotional states and behavioral histories can increase memory usage and computational demands in the game. As a result, developers often simplify these systems, which can still lead to predictable behavior loops when grounded in rigid logic.

2.1 Fuzzy Logic in NPC Decision Making

To overcome those sets of limitations where conditions are not limited to strict true and false answers, recent game development has increasingly turned to Fuzzy Logic. Unlike the industry standard frameworks that rely on hard coded output, Fuzzy Inference Systems (FIS) evaluate inputs using “degrees of truth”. This introduces a level of calculated ambiguity that closely resembles human reasoning and decision making [12][13]. Fuzzy logic allows environmental variables to exist in overlapping states. For example, rather than categorizing an enemy as strictly “Near” or “Far”, a fuzzy system might evaluate the distance as 60% “Far” and 40% “medium” simultaneously [12][13].

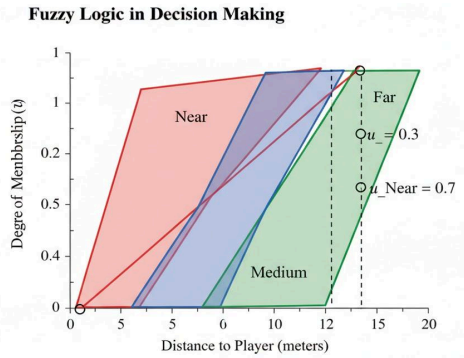

Fig 3. Fuzzy logic membership functions for player distance

This system allows NPCs to compute smoother and more natural responses, fluidly adjusting their tactical behaviors, gradually shifting from one state to another based on real-time combat, resulting in less predictable and more believable encounters [14][15][16].

2.2 Scalability Challenge of Fuzzy Logic in NPC Systems

Despite these advantages, the adoption of emotion-driven Fuzzy Logic in games is hindered by scalability constraints. Modern games generally require dozens to hundreds of active NPCs simultaneously. Simulating an independent and complex emotional brain for every single NPC that includes fuzzy sets, rule evaluations and defuzzification algorithms, introduces significant CPU overhead and major memory allocation issues, which can severely degrade game performance [17][18]. As the number of active entities grows, the memory footprint required to maintain individual fuzzy systems becomes unsustainable, which forces developers into a difficult trade-off between AI complexity and overall game performance [1].

2.3 Centralized Fuzzy Architecture for Scalable NPC Behaviour

To address the conflict between high-fidelity behavioral realism and computational efficiency, we propose a centralized Fuzzy Logic architecture as a fundamental optimization of the system design.

In a conventional implementation, every individual NPC evaluates its own distinct Fuzzy Inference System (FIS). If a game has 100 enemies, the engine must manage 100 separate memory allocations for each function, rule bases, and defuzzification processes. This will lead to massive data redundancy and severe CPU bottleneck as the number of enemies is limited [17][18].

A shared, centralized fuzzy logic architecture, drawing upon principles similar to the Flyweight design pattern [19][20], allows NPCs to be categorized by

archetypes (e.g., melee, ranged). The centralized FIS evaluates the data and returns the resulting emotional states, as well as action probabilities, to the specific NPC, which eradicates redundant memory allocation and minimizes active computations per frame, allowing complex fuzzy logic to scale efficiently alongside massive NPC counts.

2.4 Emotion Modeling Through Environmental and Player-Centric Metrics

The effectiveness of an emotion-driven AI relies on the quality of its input. According to the OCC model of emotions, affective states are valenced reactions triggered by continuous appraisal of events, agents, or objects [21][22]. In the context of video games, these appraisals translate to real-time environmental and player interaction metrics [23]. The proposed Fuzzy Logic system processes several parameters, which can be categorized into three fundamental dimensions:

1. Internal States and Resources: This dimension acts as the NPC's fundamental survival instinct evaluating its capacity to sustain and inflict damage relative to the player.

2. Spatial and Tactical Awareness: This dimension processes environmental positioning and situational advantages, allowing NPCs to react organically to the state of the battlefield.

3. Game Context: This dimension takes into account both the stage of progression as well as the difficulty level of that stage, acting as an overarching modifier for the environment.

[3] Research Methodology

This study develops a **Shared Fuzzy Logic system** for emotion-driven NPC behaviour through several stages: parameter design, architecture design, Fuzzy Inference System (FIS) development, experimental implementation, and performance evaluation.

3.1 Research Overview

A controlled game prototype is developed so that all AI approaches operate under identical conditions. The number of NPC agents is gradually increased to evaluate **frame time, memory usage, and AI processing throughput**, while behavioural validation ensures that centralized decision-making still produces believable and adaptive NPC responses [1][9].

This approach enables an objective and reproducible evaluation of both performance efficiency and behavioural realism.

3.2 System Input Parameter Design

Based on **Affective Computing** and **Appraisal Theory**, emotions arise from evaluating the agent's current state relative to its goals [21][24]. This concept is widely implemented using the OCC model as a foundation for computational emotion modelling [21].

In action-oriented games, the primary NPC goal is **survival and player elimination**, therefore the input space is limited to three key parameters:

1. Health Ratio: A relative health metric is used to evaluate NPC survivability. This allows NPCs to perform comparative appraisal that mimics human emotion, where confidence is influenced by the perceived condition of the opponent [27].

This system evaluates a relative vitality metric, with the health ratio calculated using Equation (1):

$$H_r = \frac{\text{NPC HP}}{\text{Player HP}} \tag{6}$$

2. Distance: Distance to the player represents tactical urgency and immediate threat level [4].

3. Stage: The stage parameter acts as a global difficulty modifier to maintain player engagement throughout game progression [14]. Restricting the system to three variables avoids the *curse of dimensionality* and maintains efficient fuzzy rule processing [12].

3.3 Shared Fuzzy Architecture Design

This research implements a **Centralized Shared Fuzzy Logic architecture** inspired by the **Flyweight Design Pattern** to reduce memory usage and data redundancy [19][20].

The system separates NPC data into two states:

- **Intrinsic State:** FIS, membership functions, and rule base shared across NPCs.
- **Extrinsic State:** real-time NPC data such as health ratio and distance.

This approach removes redundancy found in traditional AI architectures where each NPC maintains its own decision engine [17].

A **Central Logic Layer** processes all NPC data collectively, reducing overhead and improving scalability in multi-agent environments [7][30]. Centralized decision-making has been shown to significantly improve computational efficiency in large-scale real-time multi-agent systems [30].

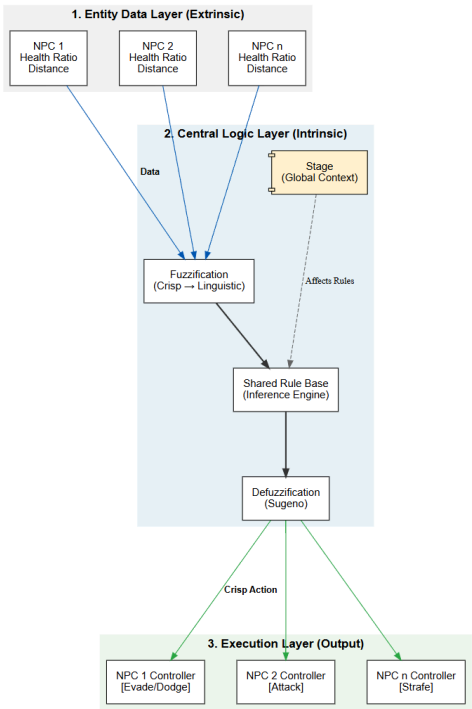

Fig. 4. Shared Fuzzy Logic Architecture Design

3.4 Fuzzy Inference System Development

The decision-making core uses a **Sugeno-type Fuzzy Inference System (FIS)** due to its lower computational cost and suitability for real-time multi-agent environments [28][29]. Sugeno FIS produces numerical outputs directly, making it well suited for adaptive control and fast decision-making systems [29].

3.4.1 Fuzzification

Real-valued gameplay inputs are converted into linguistic variables using triangular membership functions. The selected inputs are **Health Ratio**, **Distance to Player**, and **Game Stage**, representing survival condition, threat proximity, and gameplay progression.

Each input is represented using three fuzzy sets:

- Health → Low, Equal, High
- Distance → Near, Medium, Far
- Stage → Early, Mid, Late

Triangular membership functions were chosen for their computational simplicity and efficiency in real-time processing.

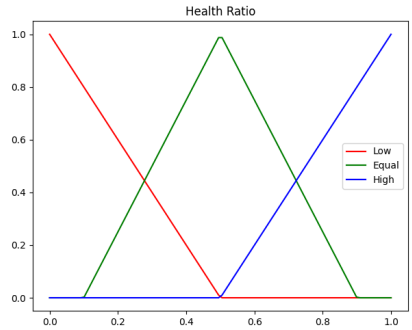

Fig. 5a. Membership functions for NPC health ratio (Low, Equal, High).

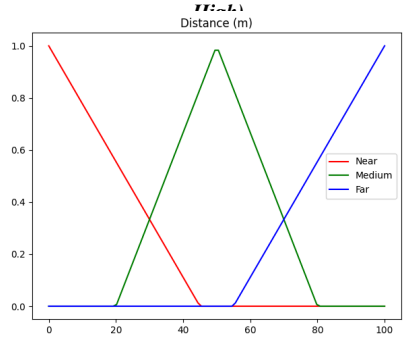

Fig. 4. Membership functions for player-to-NPC distance (Near, Medium, Far).

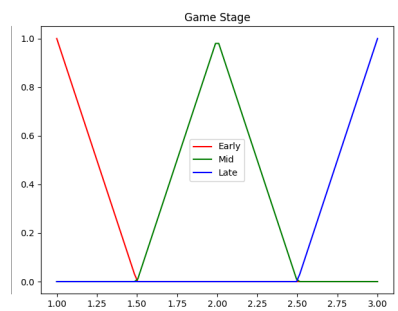

Fig. 5c. Membership functions for game stage progression (Early, Mid, Late).

3.4.2 Rule Evaluation

The fuzzy rule base consists of **27 rules**, representing all combinations of the three inputs. Fuzzy logic enables flexible decision-making based on degrees of truth rather than binary logic [13][29].

Examples of behavioural rules:

- Low health + Near → fear response
- High health + Near → aggressive response

- Balanced health + Far → neutral response
- Late stage → increased aggression

The logical AND operator is implemented using the minimum operator.

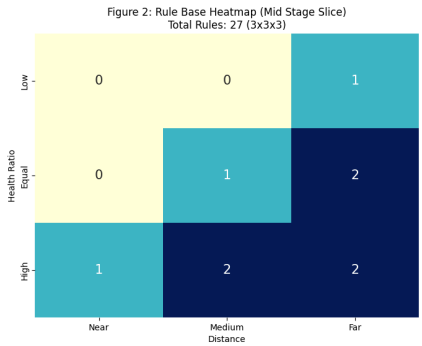

Fig. 6. Heatmap of the fuzzy rule base showing emotional output mapping for health and distance variables.

3.4.3 Sugeno Output Modeling

Unlike Mamdani FIS, the Sugeno model produces **crisp rule outputs** instead of fuzzy output sets. Each rule outputs a constant emotional value representing emotional intensity.

This significantly reduces computational overhead because the system avoids constructing and aggregating complex fuzzy output distributions.

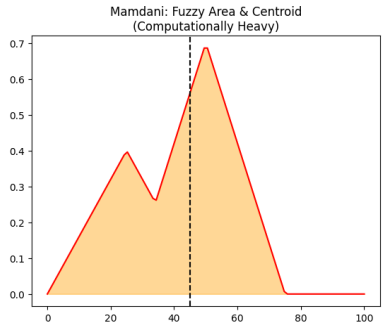

Fig. 7a. Mamdani inference process using fuzzy area aggregation and centroid defuzzification.

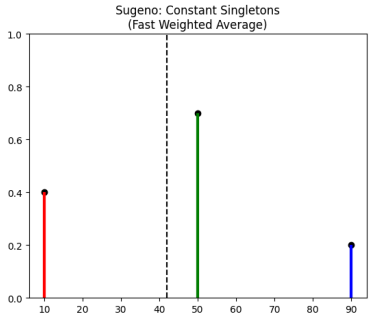

Fig. 7b. Sugeno inference process using constant singletons and weighted average for optimized computation.

3.4.4 Defuzzification

The final emotional value is computed using the **weighted average method**, which is the standard defuzzification approach in Sugeno systems. This method provides smooth transitions and is computationally efficient for real-time applications.

The resulting emotional value is mapped into behavior probabilities that determine whether the NPC performs **Attack**, **Retreat**, or **Idle** actions.

3.5 Experimental Implementation Setup

Each configuration is executed multiple times to obtain stable averaged measurements. The evaluation focuses on two research questions:

1. Does **Shared Fuzzy Logic** reduce computational overhead compared to per-NPC FIS?
2. Does **Sugeno FIS** provide better real-time performance than Mamdani while maintaining believable behaviour?

Scalability results will be visualized using comparative graphs, following established benchmarking practices for multi-agent systems [3][7][30].

Open Data

Link to Dataset or Repository

The experimental game prototype, source code, fuzzy rule base, and performance measurement scripts will be publicly available to ensure research transparency and reproducibility.

The repository will include:

- Game prototype build
- Shared Fuzzy Logic implementation
- Mamdani and Sugeno comparison configurations
- Raw experimental logs (frame time, memory usage, CPU usage)
- Graph generation scripts

Repository link:

https://github.com/TakamoriSubaru/shared-fuzzy-npc-ai

Author Contributorship

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Game designer, methodology design, software development, and experiment implementation.

Adriel Kevin Jonathan
Paper Writer & Literature Review

Jocelyn Tania Harjanto
Paper Writer & Concept designer

Kristien Margi Suryaningrum
Supervision, research direction, methodology validation, and manuscript review and editing.

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